

# JEYAKUMAR RENUJAN

Software Engineering | AI/ML | UI/UX Design | Networking

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## PROFESSIONAL SUMMARY

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Final year Computer Engineering undergraduate at the University of Jaffna with a strong foundation in Software Engineering, Artificial Intelligence, Machine Learning, and problem solving. Proficient in C++, Python, Java, and modern web technologies, with hands on experience in developing software applications and academic projects. Passionate about leveraging programming skills, analytical thinking, and emerging technologies to solve real world challenges.

## EDUCATION

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2023 - present BSc Eng at University of Jaffna  
2019 - 2022 A/L s at Mu/Puthukkudiyiruppu Central College

## RESEARCH PROJECT

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### Sentiment and Intent Detection in Code-Mixed Text

Dec 2025 - present

- Developing a multi-task NLP framework for analyzing Romanized Tamil-English code-mixed text.
- Utilizing MuRIL-based contextual representations for joint sentiment and intent classification.
- Investigating challenges in code-mixed text processing, including class imbalance and limited annotated data.
- Addressing the lack of effective NLP solutions for understanding sentiment and intent in under-resourced Romanized Tamil code-mixed communication.

## PROJECTS

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### Me Plus (Me+) - Multi-Client Task Management Platform

[GitHub](#)

Designed and developed a task management platform for freelancers to manage multiple clients, projects, tasks, deadlines, working hours, and invoices within a unified workspace. Applied user-centered design principles through questionnaire-based requirement gathering and implemented intuitive workflows, interactive dashboards, and responsive interfaces to enhance usability and productivity.

### Smart Supermarket Product Identification System

[GitHub](#)

Developed a computer vision-based product identification system using a fine-tuned YOLO11s model to detect and classify supermarket products across 17 categories. Implemented automated product counting, bounding box visualization, and statistical report generation, enabling image-based product analysis and inventory insights.

### Connectify - Real-Time Video Conferencing Application

[GitHub](#)

Developed a full-stack real-time video conferencing application using HTML, CSS, JavaScript, Node.js, Express.js, WebRTC, and Socket.IO. Implemented room-based group video calls, peer-to-peer audio/video streaming, live group chat, private messaging, and screen sharing. Integrated Socket.IO signaling for real-time event handling, participant synchronization, and meeting room management.

## AuthFlow - Authentication Based Platform

[GitHub](#)

Developed a full-stack authentication and user management platform using the MERN stack, implementing secure user registration, login, JWT-based authentication, and role-based access control. Built RESTful APIs with Node.js and Express.js, integrating bcrypt password hashing, protected routes, security middleware, and MongoDB for user data management. Designed a responsive React interface featuring user and admin dashboards, profile management, and access-controlled administrative operations.

## Intelligent Dam Safety System

[GitHub](#)

Developed an embedded IoT-based dam safety system integrating real-time water level monitoring, water level forecasting, early warning mechanisms, and automated gate control. Designed the system to monitor reservoir conditions, predict potential overflow risks, and adjust gate openings based on forecasted water levels, reducing reliance on manual intervention and supporting proactive flood risk management.

## Hostel Management System

[GitHub](#)

Developed a full-stack web-based Hostel Management System using PHP and MySQL to streamline student registration, room allocation, warden management, payments, visitor records, complaints, and check-out processes. Designed a normalized relational database with foreign key constraints to maintain data integrity, and implemented role-based dashboards for administrators, staff, and students, with persistent navigation, input validation, and secure deletion confirmations.

## Smart Load - Phone Reload System

[GitHub](#)

Developed a C++ - based mobile phone reload application with a user-friendly Graphical User Interface (GUI) using Visual Studio. Implemented core functionalities including mobile reload processing, password recovery, and input validation to handle invalid user entries and improve application usability.

## SKILLS

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|                        |                                                              |
|------------------------|--------------------------------------------------------------|
| Programming Languages  | C++, Python, Java, JavaScript                                |
| Frontend Development   | React.js, HTML, CSS, Tailwind CSS                            |
| Backend Development    | Node.js, Express.js, REST APIs                               |
| Databases              | MySQL, MongoDB                                               |
| Libraries & Frameworks | NumPy, Pandas, Scikit-learn, OpenCV, YOLO, Streamlit         |
| Tools & Platforms      | Git, GitHub, VS Code, Google Colab, Postman, MySQL Workbench |

## REFERENCES

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### Dr. (Mrs.) J. Jarachanthan

*Senior Lecturer*

*Dept. of Computer Engineering*

University of Jaffna

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